

EX NO 2: Exploration of AI Use Case: Case study on self-driving cars.

AIM

To study the application of Artificial Intelligence (AI) in self-driving cars and understand how AI technologies help vehicles perceive their surroundings, make decisions, and navigate safely without continuous human control.

THEORY

A self-driving car is an autonomous vehicle that uses AI, sensors, cameras, and computer systems to observe its environment and perform driving tasks. AI enables the vehicle to identify objects, understand road conditions, predict the movement of other vehicles and pedestrians, and select suitable driving actions.

WORKING PRINCIPLE

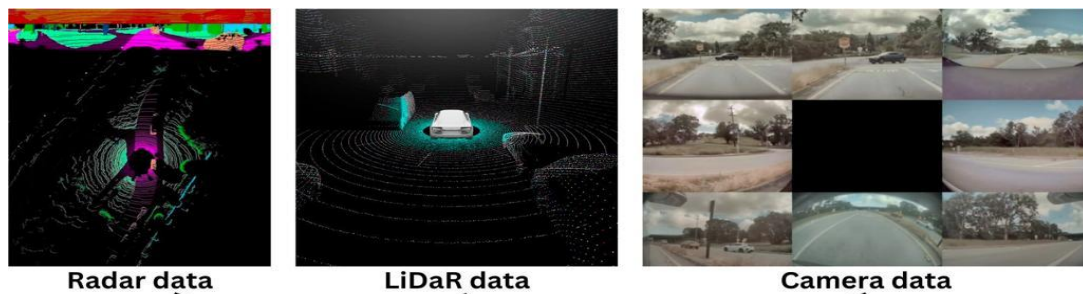
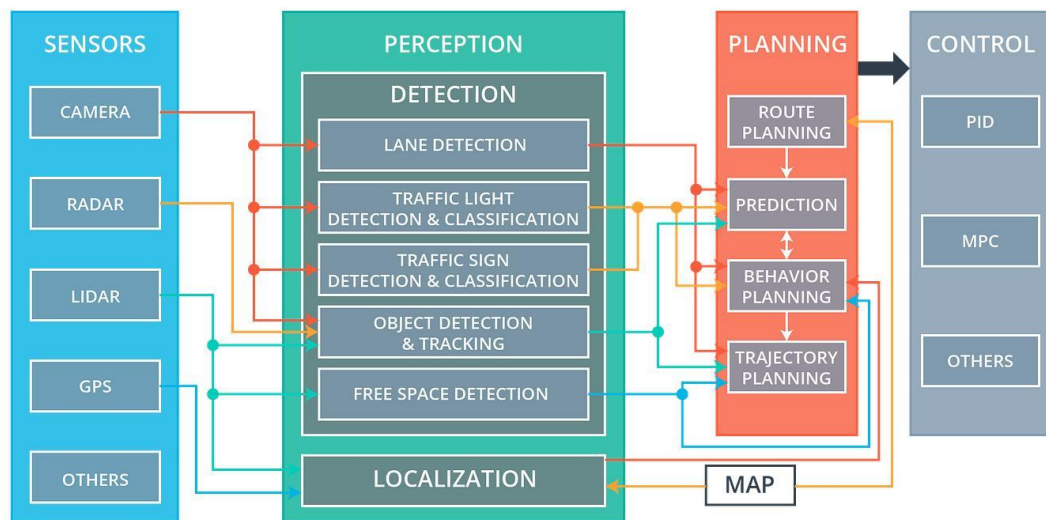
The basic working process is:

Sensors & Cameras → Perception → Object Detection → Decision Making → Path Planning → Vehicle Control

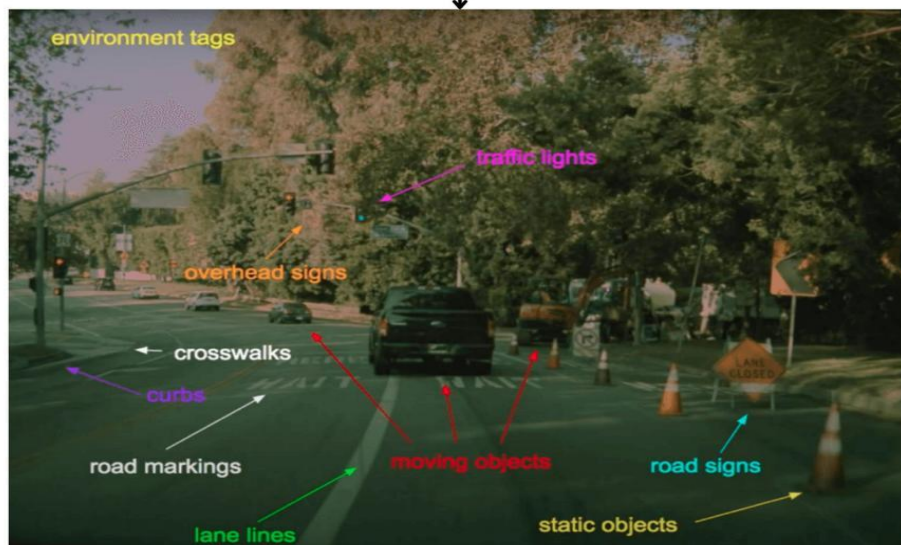
1. Sensing – Cameras, LiDAR, radar, GPS, and ultrasonic sensors collect information.
2. Perception – AI processes the collected information.
3. Object Detection – Identifies vehicles, pedestrians, traffic signs, lanes, and obstacles.
4. Prediction – Predicts the possible movement of nearby vehicles and pedestrians.
5. Decision Making – Determines actions such as stopping, turning, accelerating, or changing lanes.
6. Path Planning – Selects a safe route and driving path.
7. Control – Controls steering, braking, and acceleration.
8. Continuous Monitoring – Continuously observes the environment and updates decisions.

SAMPLE IMAGE – SELF-DRIVING CAR





Perception module



AI TECHNOLOGIES USED

- Machine Learning (ML) – Learns driving patterns from large amounts of data.

- Deep Learning (DL) – Recognizes objects, roads, traffic signs, and pedestrians.
- Computer Vision – Understands images and video captured by cameras.
- Natural Language Processing (NLP) – Supports voice-based interaction with passengers.
- Sensor Fusion – Combines information from cameras, LiDAR, radar, and other sensors.
- Reinforcement Learning – Helps improve decision-making through learning from actions and outcomes.
- Path Planning Algorithms – Determines safe and efficient driving paths.

CHARACTERISTICS

1. Autonomous Driving :Performs driving tasks with minimal human intervention.
2. Object Detection : Detects vehicles, pedestrians, cyclists, and obstacles.
3. Lane Detection :Identifies road lanes and lane boundaries.
4. Traffic Sign Recognition :Recognizes traffic signs and signals.
5. Obstacle Avoidance: Detects obstacles and takes appropriate action.
6. Automatic Braking :Applies brakes when a collision risk is detected.
7. Path Planning:Selects a suitable and safe route.
8. Real-Time Decision Making:Makes driving decisions based on current conditions.
9. Navigation :Uses GPS and digital maps to reach the destination.
10. Driver Assistance : Provides features such as adaptive cruise control and lane keeping.

RELATED AI TOOLS / TECHNOLOGIES

1. Machine Learning (ML) :Learns driving patterns from collected data.
2. Deep Learning (DL) :Helps recognize objects, roads, vehicles, and pedestrians.
3. Computer Vision : Analyzes camera images and identifies road conditions.
4. LiDAR Processing: Creates a 3D representation of the surrounding environment.
5. Sensor Fusion |: Combines information from cameras, LiDAR, radar, and other sensors.
6. Reinforcement Learning : Helps the vehicle learn and improve driving decisions.
7. Path Planning : Determines a safe and efficient route for the vehicle.
8. Predictive Models : Predicts the movement of nearby vehicles, pedestrians, and cyclists.

CONCLUSION

Self-driving cars are an important real-world application of AI. They combine machine learning, deep learning, computer vision, sensor fusion, and path planning to perceive the environment and make driving decisions. AI-based autonomous vehicles can improve transportation safety, efficiency, and convenience, although challenges such as safety, reliability, cost, cybersecurity, and ethical decision-making remain.